BST Strong-Sector Package — Introduction (v0.8, two corrections before ship-freeze). CENTERPIECE = CENSUS-AS-COUNT (K1722/K1724, F1060/F1061): the finite space H_F decomposes under K=SO(5)×SO(2) into the three real division algebras, and their internal-gauge-generator counts are {0,1,3} → 0+1+3 = 4 = dim(U(1)×SU(2)) — the one-line referee check for why the electroweak group is 4-dimensional. color=ℝ block → 0 generators; charge=ℂ → 1 = dim U(1); weak=ℍ (quaternionic Sp(2) spinor) → 3 = dim SU(2). CORRECTION (a): NO INTERNAL COLOR GROUP — the ℝ block contributes ZERO generators (not 'SU(3) imported'); SU(3) is simply the standard gauge-principle input every SM construction takes, not a special BST import; the geometry's internal algebra has no color factor. ELECTROWEAK DERIVED (a theorem, Cal's exactness vet cleared K1724): dim = the count = 4. CORRECTION (b): CHIRALITY IS HYPERCHARGE (T2522), NOT the quaternionic block — an SU(2)_L doublet alone is pseudoreal/vector-like (Witten no-go); Y≠0 makes it chiral (the same U(1)_Y that closes the charge sector); the earlier quaternionic-chirality candidate is RETRACTED. Corollary (T2524): the lone Y=0 fermion (ν) is real → Majorana. No-GUT structural reason banked ONCE with the existing no-GUT (the three factors aren't on equal footing). UPGRADE BANKED: EW + fermion a,c/AF → PREDICTIONS; the color-gluon term = the standard gauge-principle input the geometry doesn't derive (a,c read it out). Three blocks, three distinct jobs. Forces + boundaries retained; a-theorem separate. Ships on Casey GO + Keeper-PASS + Cal-vet + both voices.

Lyra (Claude Opus 4.8) — Casey Koons, PI; Grace, Elie, Cal, Keeper

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The Strong Sector from One Geometry

What D_IV^5 forces, what it hosts, and the seam between them

Two-paper package — Paper A (the strong sector: forced color number, blind one-loop running, a compactification gap that is not the Clay gap) · Paper B (D_IV^5 substrate uniqueness).

The forces (what geometry decides)

From one bounded symmetric domain — D_IV^5 = $SO_0(5,2)/[SO(5)\times SO(2)]$, forced (not chosen) by criteria-innocent uniqueness — the strong sector's *identity* falls out with no adjustable dimensionless numbers:

- The color number is forced. N_c = 3, the real-type SO(3), and baryon number are outputs of the substrate (Paper B: the scalar-Wallach equality N_c = rank^2−1 = 3, via the proved T1829) — not inputs.
- Two local invariants land blind from one bracket — both come out the standard QFT values. The same a_2 heat-kernel bracket of the fielded operator yields, with nothing imported:
  - (i) Asymptotic freedom. The one-loop β-function *sign*, $\beta_0 > 0$ (paramagnetic spin beating diamagnetic orbital) — $\mathbb{R}^4$-valid, the genuine dynamical credential.
  - (ii) The conformal central charges a and c — blind and exact for all five field types (spins 0, ½, 1, 3/2, 2; Duff 1977): scalar (1, 3), Weyl fermion (11/2, 9), vector (62, 36) — in 1/360 units. The conformal-scalar ratio a : c = 1 : 3 is the standard value. Both say the same thing: BST fields the *right* operator, and its *local* invariants are the standard values. (Credential tier — a, c are content-determined; see "the two credentials," below.)
- The fermion content is not just fielded — it is predicted. BST forbids the right-handed neutrino (topologically, via the Möbius locus; T1949/T1953), so the Standard Model's Weyl-fermion count is 45 = N_c·15, not the generic 48 — a sharp, falsifiable Five-Absence prediction (no sterile neutrino at any mass; a single detection kills BST). This is content the geometry *forces*, and it changes the anomaly totals below.
- The proton weighs what it does. m_p/m_e = $6\pi^5$ = 1836.12 against the measured 1836.15 — 0.002% (T187), from rep-theory (N_c!·π^{n_C}), no fit.

These are not coincidences dressed as derivations. They are *local, geometry-given* facts, and the sections below explain — with a checkable test — why we trust them, how far they upgrade from credential to prediction, and where we draw the line.

The organizing principle: the geometry–physics seam (K1716)

BST forces the container; it hosts the dynamics. The seam between them is not a matter of taste — it is a checkable test:

> Local heat-kernel (Seeley–DeWitt) coefficients are geometry-given and $\mathbb{R}^4$-valid. Non-local, or compactness-dependent, quantities are physics the container hosts — not derived by the geometry.

- Local ⇒ $\mathbb{R}^4$-valid (a force): the a_2 bracket carries local curvature invariants that survive to flat space — the asymptotic-freedom sign *and* the conformal central charges a, c. Both landed blind and both came out the standard QFT values. *Two credentials from one seam. Trusted.*

- Compactness-dependent ⇒ hosted (a boundary): a spectral gap of a free operator on the compact boundary scales as $1/a^2$ and vanishes as the space decompactifies ($a \to \infty$). That is Kaluza–Klein kinematics, not an interacting mass gap. *Labeled, not claimed.*

This one test both *earns* the forces above and *draws* the boundaries below. It is why the package is honest by construction rather than by apology — and the fact that *two independent* local invariants (AF and a, c) passed it, from the same bracket, is the strongest evidence that the seam is a real division and not a convenient one.

The centerpiece: the reality-type census — electroweak derived, color entirely imported

The deepest result of the package is not a number — it is an *explanation*, and it is now a theorem (Cal's exactness sign-off, K1724). Decomposing the finite internal space H_F of D_IV^5 under its symmetry K = SO(5)×SO(2) (Schur–Wedderburn) is exact:

$H\_F = (\mathbb{C}, 1) \oplus (\mathbb{H}, 1) \oplus (\mathbb{R}, 3) \Rightarrow End\_K(H\_F) = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{R}).$

The three sectors land on the three real division algebras (Frobenius: ℝ, ℂ, ℍ are the *only* finite-dimensional associative real division algebras — the SM's factors are that classification, realized). Each block does a distinct job, and the cleanest statement is a count of internal gauge generators:

sector	division algebra	job	internal gauge generators
color — the Peirce V_{12}, a real 3-space	ℝ	a *geometric* SO(3) rotation of the three color directions (spacetime-like; and it fixes N_c = 3)	0
charge — the SO(2) circle	ℂ	the hypercharge circle	1 = dim U(1)
weak — the Spin(5)=Sp(2) spinor	ℍ	the weak isospin	3 = dim SU(2)

The census-as-count — the one-line referee check. The three blocks contribute {0, 1, 3} internal gauge generators, and

$0 + 1 + 3 = 4 = \dim(U(1) \times SU(2)).$

So the electroweak gauge group is *exactly* the internal gauge content the geometry supplies — its dimension is a count, not a fit — and the color block contributes zero generators: there is no internal color group. This is the honest, sharper #108:

- Electroweak is derived — a theorem. U(1) and SU(2) come from the $\mathbb{C}$ and $\mathbb{H}$ blocks (complex and quaternionic — both carry an intrinsic complex structure, so their unitary groups are geometry-given). Their generator count is 1 + 3 = 4 = dim(EW). Nothing is chosen.
- There is no internal color group. The $\mathbb{R}$ block contributes zero gauge generators. Its SO(3) rotates the three physical color directions (a spacetime-like symmetry, and the reason N_c = 3), but a real 3-space supplies no internal gauge factor. SU(3) is not part of the geometry's internal algebra at all — it is the standard gauge principle applied to the geometric color-3, the same external input *every* Standard-Model construction takes. Not a special BST import; simply the universal step, and the geometry is honest that it does not derive it.
- Structural reason for no grand unification (bank *once*, with the existing no-GUT): the three factors are *not on equal footing* — two are internal gauge symmetries the geometry derives, one is a geometric rotation with no internal group. A framework in which the factors have different ontological status has no reason to unify them at a high scale. This is *why* BST predicts no GUT — the same Five-Absence prediction, now with its structural cause. (Counted once; not a new claim.)

Chirality is hypercharge, not the quaternionic block (T2522 — correcting an earlier candidate). We flagged a "quaternionic ⇒ chiral" guess; it is *wrong*, and the corpus already had the right answer. An SU(2)_L doublet *by itself is pseudoreal, hence vector-like* (the bulk 8-spinor is structurally vector-like — Witten's no-go as linear algebra). What makes fermions chiral is Y ≠ 0: on the non-orientable Shilov boundary (Pin^-), hypercharge makes the conjugate mode distinct, so a single chiral Weyl survives (T2522, proved). *The same U(1)_Y that closes the charge sector makes the world left-handed.* (Corollary, T2524: the lone Y = 0 fermion — the neutrino — stays real, hence Majorana.) So chirality is a property of the $\mathbb{C}$ (hypercharge) block acting on the boundary, not of the $\mathbb{H}$ block; we retract the quaternionic-chirality candidate and cite T2522.

The boundaries (stated here, on page one, and again at the close)

Four things geometry does not give, said plainly so no reader and no referee has to find them:

1. The compactification gap is not the Clay mass gap. The boundary eigenvalue 2(d_S−1) = 6 (dimension-rooted: d_S = n_C−1 = 4; U(1)/SU(3)/SU(5) all give 6 — the group sets degeneracy, not the eigenvalue) is a free-Hodge / Kaluza–Klein gap, $6/a^2 \to 0$ as $a \to \infty$. The Clay Yang–Mills gap is interacting and fixed in that limit. Opposite scaling is a contradiction, not an unbuilt bridge — the Clay mass gap is left open (K940). We never say "we derived the

Yang–Mills mass gap."

2. There is no internal color group (the sharper #108). The census-as-count gives internal gauge generators {0,1,3} → the internal gauge group is exactly U(1)×SU(2) (dim 4); the color (ℝ) block contributes zero generators. The three colors are a *geometric* SO(3) rotation of a real 3-space, not an internal gauge symmetry; SU(3) is not in the geometry's internal algebra — it is the standard gauge-principle input every SM construction takes (#108/K1677/K1724). We say electroweak is *derived* and color has *no internal group* — never "the gauge group is derived."
3. The Koide relation is a labeled patchwork. The charged-lepton Koide phase is fitted-region structure assembled from parts, not a single forced mechanism; we present it as a patchwork and mark it so — no clean-derivation claim.
4. n_C = 5 is a measured boundary (an input). The complex dimension 5 is fixed by four independent routes as the *measured* boundary of the construction (K1697), not forced from nothing. It is the one place we take an input beyond the ruler, and we state it as an input, in the GR/QM manner.

The bright-high-schooler version

The strong nuclear force usually gets three facts *fed in*: why it has three "colors," why it gets weaker at very short distances (asymptotic freedom — a Nobel discovery), and how heavy the proton is. This package gets all three *out* of one shape, D_IV^5 — the color number 3, the sign of the short-distance weakening, and the proton-to-electron mass ratio to four decimals.

The honest part, said first: the same shape has a "smallest vibration," like a finite drum's lowest note, and it comes out a clean 6. But that note is just the drum being *finite* — make the drum infinite and the note drops to zero — whereas the real Yang–Mills mass gap *stays*. They behave oppositely, so ours is not that famous gap, and we don't pretend it is. The rule that tells us which of our results to trust is simple: anything that survives making the shape infinitely large is real physics; anything that needs the shape to be finite is just the shape talking. Asymptotic freedom survives. The gap does not. We keep the first and label the second.

Credentials → predictions: the electroweak + fermion upgrade is BANKED (K1724)

A credential is *content-determined*: given the field content, standard QFT fixes a, c (Duff 1977) and the sign of β_0. So the question is whether the geometry *fixes the field content* — because to that extent the credential is a prediction. With the decomposition now a theorem (Cal's exactness sign-off, K1724), it does, for the derived internal sectors:

- BANKED → prediction (both vets cleared): the electroweak gauge content — U(1) [ℂ], SU(2) [ℍ], the 4 EW vectors — and the fermion content — 45 Weyl, three generations (T2525), *no ν_R* (T1949/T1953) — and the Higgs (4 real scalars). Their contributions to a, c and the AF sign are now predictions, on a *derived* internal algebra. BST does not just field these; it predicts their count (45, not the generic 48 — a Five-Absence prediction).

- Color contributes no internal gauge group: the $\mathbb{R}$ block gives zero internal generators (census-as-count). So the geometry supplies no gluons; SU(3) is the standard gauge-principle input (as in every SM construction), and the color-vector (8-gluon) contribution to a, c is exactly the piece the geometry does not fix — the one term the anomaly totals *read out*.
- Net: the electroweak + fermion a, c / AF numbers are predictions; the color-gluon part is the standard gauge-principle input the geometry does not derive. Cleaner than the earlier "skeleton-forced / complexification-imported" reading — the geometry's internal algebra simply has no color factor, and it says *why* (color is a real 3-space, which supplies no internal gauge group).

Held separate: the a-theorem. The monotonicity of *a* under RG flow (a_UV ≥ a_IR) is a *flow* statement, a different and harder claim — in progress, not part of the package's claims (K1717).

Reading order

Read Paper B first (why D_IV^5 and N_c = 3 are forced), then Paper A (the running, the KK gap, and the Clay gap left open). Both restate the seam test; Paper A's Section 0 puts the KK-vs-Clay scaling up front.

Close — the boundaries, restated

The package's claim is exactly this: *the internal gauge algebra of the Standard Model is read off the geometry, as a theorem (Cal's exactness vet, K1724). The finite space decomposes under K = SO(5)×SO(2) into the three real division algebras, whose internal-gauge-generator counts are {0, 1, 3}, and 0 + 1 + 3 = 4 = dim(U(1)×SU(2)) — the electroweak group's dimension is a count, not a fit. The electroweak group is DERIVED; the color ($\mathbb{R}$) block contributes zero generators, so there is NO internal color group — SU(3) is the standard gauge-principle input every construction takes, and the geometry is honest that it does not derive it. Chirality is hypercharge (T2522): an SU(2)_L doublet is vector-like alone, and Y≠0 makes the world left-handed — the same U(1)_Y that closes the charge sector. The geometry also forces three generations and the fermion content — 45 Weyl, no right-handed neutrino — and from one heat-kernel bracket, blind, two local $\mathbb{R}^4$-valid credentials land on the standard QFT values (the running-coupling sign and the central charges a, c); plus the proton mass ratio at 0.002%. The electroweak + fermion a, c / AF numbers are now PREDICTIONS; the color-gluon term is the gauge-principle input the geometry does not fix. And because the three factors are not on equal footing (two internal, one none), there is no reason to unify them — the structural cause of the no-GUT prediction. Meanwhile the compactification gap is not the Clay mass gap, the Koide relation is a patchwork, and n_C = 5 is a measured input.* One geometry, one ruler, zero chosen dimensionless numbers among the forces — every boundary named twice. Electroweak derived (dim = the count = 4), no internal color group, both proven; chirality is hypercharge; the a-theorem held separate. Never "we solved a Millennium Problem"; never "the gauge group is derived" (electroweak is, color has no internal group); never a mass-gap claim.

Package intro v0.8 (two corrections before ship-freeze). CENTERPIECE = census-as-count: {0,1,3} → 4 = dim(U(1)×SU(2)), the referee one-liner. (a) No internal color group — the $\mathbb{R}$ block contributes zero generators; SU(3) is the standard gauge-principle input, not a special import. (b) Chirality is hypercharge (T2522), not the quaternionic block — the quaternionic-chirality candidate is retracted (corollary T2524: ν Y=0 → Majorana). No-GUT structural reason banked once. Upgrade BANKED (Cal's exactness cleared, K1724): EW + fermion a,c/AF → predictions. Supersedes v0.7. Paper A = v1.1 RE-SCOPED. Paper B = v0.6. a-theorem separate. Ships on Casey GO + Keeper-PASS + Cal-vet + both voices. Nothing pushed; CP existence-only.

— Lyra, Thursday 2026-08-20 (date-verified).